awalla approached the Drugs Controller of India to begin clinical trials in India around the same time that Covaxin had sought permission to do so.

Human clinical trials involve three phases—in the first phase, a limited number of people are given the vaccine to test its safety and ensure that there is no adverse reaction. If the first phase is cleared, then a larger group of volunteers are injected to understand the immune response. If the phase 2 trials are successful, an even larger number (26,000 in Covaxin's case) of people are randomly assigned to receive the vaccine or given a placebo and are evaluated for how well the vaccine protects individuals and monitored for any adverse reactions.

Usually, these trials take several years before the vaccines are cleared for use. But the central government stepped in to fast-track the process without compromising on the regulatory rigour and integrity. The Department of Biotechnology facilitated multiple clinical trials of various vaccine candidates. The Drugs Controller also collapsed the first two phases into one and formed committees to speedily evaluate the results using international guidelines developed by the WHO.

The clinical trials saw both Ella and Poonawalla put on trial by the media. Ella recalls that during the Covaxin trials one of the volunteers in Madhya Pradesh reportedly died and protests broke out across the state. Since Ella had outsourced the trial process to a multinational company to observe best practice norms, he had to wait anxiously for their findings. The company reported back that the volunteer was given a placebo and not the actual vaccine. Ella was relieved. Such was his worry that even before the human trials began, he was the first one to take Covaxin to ensure that it was safe. Only then did his senior colleagues follow. He says, "It is important we should be able to take it ourselves before we test it on the general public." Ella recalls that when he developed the vaccine for the Rotavirus, which causes diarrhoea in children, he first tested it on his young grandson after it got cleared.

Poonawalla also found the development and trial period for his vaccine a challenging time given the intense media scrutiny. He recalls, "We took it in our stride. The only thing driving me was that every month we lost could result in more lives being lost." When the trials were almost complete, his 12-year-old son Cyrus asked him, "What will happen if Covishield doesn't work against this virus...what are you going to do?" Adar told him, "I will have to look for another job



INTERVIEW

"Covaxin was like India's nuclear moment"

Dr Krishna Ella, chairman and managing director of Bharat Biotech, speaks to Group Editorial Director **Raj Chengappa** on India's vaccine prowess and the need for young entrepreneurs to innovate. Excerpts:

Q. Why did you decide to take the indigenous route to make the Covid-19 vaccine?

For me, it was equal to India's nuclear moment—in the past, only powerful countries had the sophisticated technology to make atom bombs, but India demonstrated that we could make them indigenously. I saw this as a similar opportunity for a new generation of entrepreneurs to demonstrate that India could be a world leader in the field of vaccinations.

Q. It takes anything from 5 to 10 years to develop a vaccine. How did you do it in a year?

The rotavirus vaccine took 17 years in the making, typhoid 10 years. So how did we succeed in making the Covid vaccine in a year? It was because there was a global requirement, so both the regulatory system and the government proactively pushed for it. This is where public-private partnership comes into play. Apart from our own development,